Praneeth Reddy Ankey

+91 8179141580 | apraneethreddy20891a0502@gmail.com | linkedin.com/in/praneeth-reddy-ankey | Hyderabad, India

Professional Summary

Full Stack & GenAI Engineer with 1+ year of production experience building agentic RAG systems, AI-powered pipelines, and real-time web applications using Python, React.js, TypeScript, and FastAPI. Skilled in LLM integration (Llama 4, Groq SDK, OpenAI-compatible APIs, Ollama), vector databases (Supabase pgvector, ChromaDB), prompt engineering, and automated document-processing workflows. Proven delivery across defence (DRDO) and enterprise environments. Passionate about fusing generative AI with scalable full-stack development to ship user-centric products.

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, SQL, Java, C, C++
AI / GenAI: Llama 4, Groq SDK, OpenAI API, Ollama, Pydantic AI, LangChain, LlamaIndex, Crawl4AI, PaddleOCR, Pix2Tex, Nanonets OCR, sentence-transformers, Prompt Engineering, Text Embeddings, Chunking Strategies, RAG Pipelines, Agentic AI, AI Summarization Workflows
Frontend: React 19, React.js, Next.js, TanStack (Start, Router, Query), Electron, Vite, Tailwind CSS, Vega-Lite, HTML5, CSS3
Backend: FastAPI, Node.js, RESTful APIs, WebSockets, JWT / OAuth 2.0, Clean (Hexagonal) Architecture, Dependency Injection, SQLAlchemy, Alembic, Asynchronous / Event-Driven Programming
Databases / Vector Stores: Supabase pgvector, ChromaDB, PostgreSQL, SQLite, SQL, Pinecone (familiar)
Data Processing: Arquero, PyMuPDF, OpenCV, openpyxl, Hugging Face Transformers, watchdog (Python), Web Scraping
Tools & DevOps: Docker, GitHub Actions (CI/CD), Git, AWS S3 / GCP Cloud Run, Microsoft Teams API, VS Code
Concepts: Semantic Search, Hierarchical Data Modelling, Multi-Agent Systems, Streaming LLM Responses, Vector Search

Work Experience

Defence Research & Development Organisation (DRDO) — Hyderabad, India
Project Intern — Full Stack Developer — *Nov 2023 – May 2024 (213 Days)*

- Led end-to-end development of a Live Missile Data Simulation system, owning both front-end and back-end modules across the full 213-day project lifecycle.
- Built real-time missile trajectory dashboards in React.js with dynamic, responsive UI components rendering live simulation data streams.
- Developed Python/FastAPI backend services for high-throughput data ingestion, processing, and automated mission analysis report generation, ensuring reliable server-client communication.
- Designed a 4-level hierarchical data model (Main Category → Sub Category → Time-based grouping → Metrics) for the NOC Dashboard, enabling faster graph rendering and efficient filtering of large-scale Excel-sourced datasets.

Cognitbotz — Hyderabad, India
Software Development Intern — *170 Days*

- Designed and delivered Meet-Ops, an autonomous meeting bot that joins Microsoft Teams calls, captures transcripts, and surfaces AI-driven insights via a React.js web dashboard.
- Built an end-to-end AI summarization pipeline transforming raw transcripts into actionable meeting insights, integrated with backend REST APIs.
- Deployed cloud-integrated services using Docker, PostgreSQL, and Microsoft Teams API; implemented JWT-based authentication for secure persistent data storage.
- Set up a CI/CD pipeline using GitHub Actions to automate containerised deployments, reducing manual deployment overhead.

Projects

OpenViz — Generative AI Analytics Platform — *React 19, TypeScript, Vega-Lite, Llama 4, Groq SDK, Arquero*

- Architected a React 19 & Vega-Lite visualization engine with a hybrid drag-and-drop / natural-language interface powered by Llama 4 via Groq SDK, enabling prompt-driven chart generation.
- Implemented a client-side RAG pipeline with Arquero for instant data profiling and context-aware chart suggestions, with text embeddings and chunking strategies ensuring zero-latency, privacy-preserving interactions.
- Applied prompt engineering techniques to optimise LLM output structure for reliable Vega-Lite spec generation.

SiLens AI — AI-Powered STEM Learning Platform — *FastAPI, React, TypeScript, Groq LLM, PaddleOCR, Pix2Tex, SQLite/PostgreSQL*

- Architected an end-to-end document parsing and ingestion pipeline, integrating OpenCV preprocessing (denoising, deskewing) with PaddleOCR and Pix2Tex to extract plain text and compile math equations into clean LaTeX.

- Built a FastAPI backend using Clean (Hexagonal) Architecture with dependency injection, supporting hot-swappable LLM providers (Groq/Llama-3) and caching generated educational outputs to reduce API latency and compute costs.
- Designed a dynamic, state-safe frontend in React and TypeScript utilizing TanStack Start, Router, and Query, enabling real-time document upload, pipeline polling, interactive quizzes, and contextual document-grounded Q&A chat.
- Configured database schema migrations with Alembic and SQLAlchemy, implementing custom JSON-to-JSONB mapping variants to support seamless local development on SQLite and high-volume data storage on PostgreSQL.

AI File Explorer — Local AI-Powered File Indexing & Search Platform *FastAPI, React, TypeScript, Electron, Ollama, Groq, ChromaDB, PyMuPDF*

- Architected an asynchronous, event-driven file watch and ingestion pipeline utilizing Python's `watchdog` library to monitor local directories in real-time, parsing multi-format documents (PDFs, Markdown, source code) with `PyMuPDF` and executing chunking/embedding operations in a background `ThreadPoolExecutor` to keep FastAPI non-blocking.
- Built a local vector search engine using `sentence-transformers` (`all-MiniLM-L6-v2`) to generate document embeddings, persisting them in a local `ChromaDB` vector database, with bounded index retrieval constraints to prevent query crashes during semantic RAG.
- Developed a hybrid LLM execution layer supporting local `Ollama` (`llama3.2`) and cloud-based `Groq` API endpoints, enabling real-time token streaming and agentic tool-use to trigger local file-system actions from natural language queries.
- Designed a type-safe desktop application packaging a React/TypeScript frontend inside an `Electron` shell, using TanStack (Start, Router, Query), a unified Vite dev proxy to resolve CORS locally, and hydration-safe client-side state hooks.

Meet-Ops — Meeting Automation System *Microsoft Teams API, React.js, TypeScript, Docker, PostgreSQL*

- Built an autonomous meeting assistant bot that joins Teams calls, captures and stores transcripts, and surfaces AI-generated summaries through a connected React dashboard.
- Integrated backend REST APIs with cloud services (Docker Hub, AWS); implemented AI summarisation with Hugging Face Transformers for actionable insights.

Live Missile Trajectory Visualisation & Report Generation *React.js, Python, FastAPI, WebSockets*

- Developed a real-time missile trajectory visualisation system for DRDO, processing live simulation data and rendering it dynamically on a React.js front end via FastAPI + WebSocket services.
- Built Python backend pipelines for automated data ingestion, processing, and mission analysis report generation.

Education

Vignan Institute of Technology and Science Hyderabad, India
Bachelor of Technology — Computer Science and Engineering (CSE) *Graduated 2024*

- CGPA: 7.5 / 10

Telangana State Board Telangana, India
12th Grade (Intermediate) *2020*

- Score: 84.5%

Certifications

PCAP — Programming Essentials in Python (Cisco / Python Institute)
PSAP — Programming Essentials in Python – Advanced (Python Institute)
CPA — Programming Essentials in C++ (Cisco / C Institute)
CLA — Programming Essentials in C (Cisco / C Institute)